

DIGITAL PUBLIC SAFETY

PRAHARI

Stop the scam before the money moves.

Privacy-first live scam intervention and fraud-ring intelligence

PROJECT OUTCOME A working web prototype that detects digital-arrest coercion while a call unfolds, interrupts before payment, and converts only citizen-approved evidence into ranked fraud-ring intelligence.

Hackathon Project Report | 22 July 2026

Citizen: localhost:3000/check | Officer command: localhost:3000/command

Executive Summary

PRAHARI is a privacy-first digital public-safety prototype designed for the decisive moment in a digital-arrest scam: before the victim transfers money. It combines live transcript capture, on-device deterministic risk detection, optional structured AI classification, consent-controlled evidence sharing, and graph-based fraud-ring discovery in one working web system.

The citizen experience issues an immediate, plain-language intervention when scam pressure becomes critical. SAFE content stays on the device. For risky content, the user chooses whether to keep the result private or share selected evidence, a redacted transcript, or the full transcript. Approved identifiers are persisted with an audit trail, ingested into Neo4j, and surfaced to officers as ranked rings and hub infrastructure.

WHY IT MATTERS PRAHARI identifies the coercion pattern while it develops and links consented evidence across reports so both citizens and officers can act earlier.

1. Problem Statement

Digital-arrest scams combine authority impersonation, fabricated legal evidence, isolation, urgency, and a final payment demand. Victims are pressured to remain on a call, avoid family members, and transfer funds to a so-called safe account. Conventional reporting is often retrospective, after the transfer has occurred, while individual reports remain disconnected from the shared phone numbers, UPI IDs, and bank accounts used across a campaign.

Design challenge

- Warn a citizen during the scam without requiring technical knowledge.
- Keep content local unless risk is found and the citizen explicitly consents.
- Remain useful when the optional LLM or network is unavailable.
- Turn minimal approved evidence into operational intelligence without claiming a legal determination.

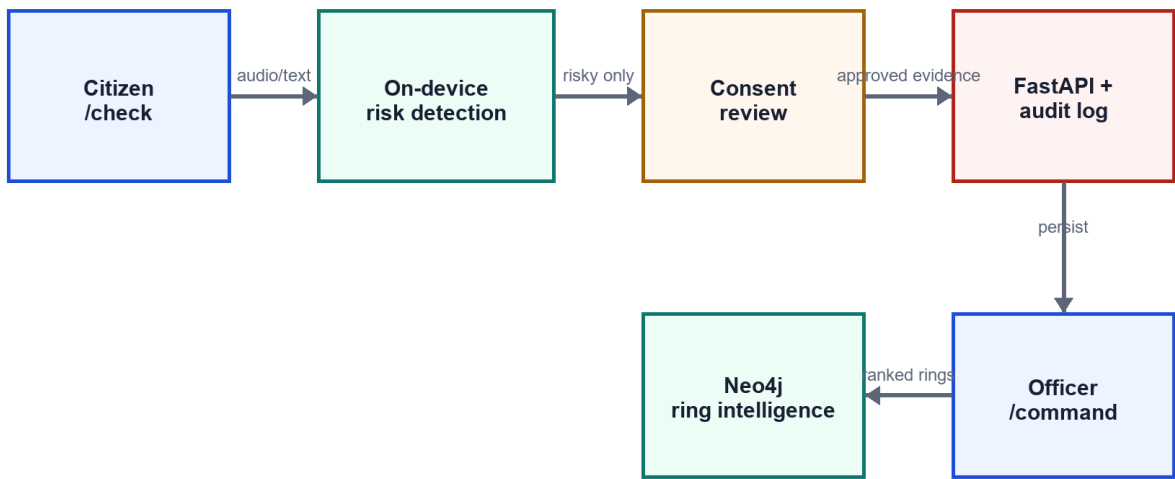
2. Proposed Solution

PRAHARI closes the loop from citizen protection to officer prioritization. The web interface accepts live speech, pasted messages, or text files. A local deterministic pass detects scam phases and fuses their signals into SAFE, SUSPICIOUS, or CRITICAL. Critical results trigger a full-screen intervention with a direct 1930 action and a trusted-contact path. Only after consent does the backend persist evidence, create an audit record, and search for shared infrastructure in the graph.

Capability	Implemented behavior
Live protection	Interim browser speech and segmented local Whisper feed continuous on-device checks.
Five-phase detection	Hook, Authority, Fabricated Evidence, Isolation, and Drain update as the transcript grows.
Privacy gate	SAFE content is never sent; risky content exposes explicit evidence-scope choices.
Fraud intelligence	Phone numbers, UPI IDs, bank accounts, and agencies collapse into shared graph nodes.
Officer command	Incident queue, KPIs, ranked rings, hubs, graph exploration, and evidence packages.

3. System Architecture

PRAHARI: privacy-first intervention to fraud intelligence



SAFE content remains local. Nothing is shared until the citizen approves a specific evidence scope.

Figure 1. Device-first analysis, consent boundary, persistence, graph intelligence, and officer workflow.

Technology stack

Layer	Technology	Role
Citizen web	Next.js 16, React 19, Tailwind	Live check, evidence review, intervention, and complaint drafting
Local speech	SpeechRecognition + Transformers.js	Instant capture and private Web Worker Whisper
Detection	Rules + optional structured LLM	Phases, entities, spans, fusion, and offline fallback
Application API	FastAPI, SQLAlchemy, Alembic	Consent, persistence, audit, lifecycle, and dashboards
Data	PostgreSQL + Neo4j GDS	Incident records and connected fraud infrastructure
Deployment	Docker Compose	Repeatable four-service demo environment

4. Citizen Experience

Common-man-first interaction

The default route asks one direct question: Is this call or message a scam? The primary option is Protect me during a call. Controls use plain language, large actions, visible privacy status, and one critical instruction: Stop. Do not send money. Optional reporting remains collapsed until needed.

Hybrid speech capture

- Fast mode uses browser SpeechRecognition and visibly warns that the browser vendor may process audio online.
- Private mode downloads Whisper once, then transcribes short segments in a Web Worker with WebGPU or WebAssembly.
- The selector includes English and all 22 scheduled Indian languages; low-resource offline cases are labeled best-effort.

- The live demo follows the local analysis path and demonstrates intervention before the synthetic payment request.

Live scam escalation and intervention

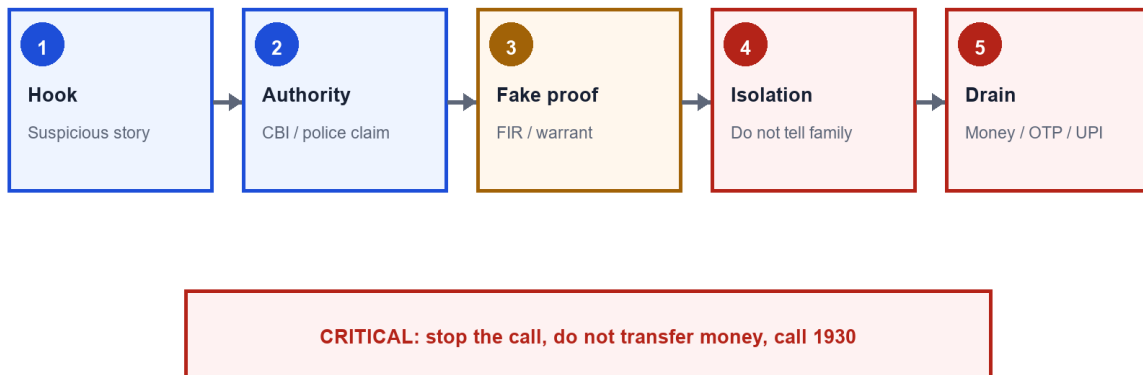


Figure 2. Progressive phase detection and citizen intervention.

5. AI/ML Detection Pipeline

Mit's module provides the detection core. A deterministic rule engine identifies authority, isolation, payment, fabricated legal evidence, urgency, and scam-hook patterns. The optional structured classifier adds phase detection, entity extraction, evidence spans, confidence, and plain-language reasoning. Calibrated fusion produces the final risk label. When the classifier is unavailable, the same API returns an offline safety verdict instead of failing.

Stage	Output	Operational value
Rules	Matched categories, evidence, score	Deterministic, fast, explainable, and offline
Structured classifier	Phases, entities, spans, confidence	Richer reasoning when a model is available
Fusion	SAFE / SUSPICIOUS / CRITICAL	Consistent intervention and persistence thresholds
Incremental analysis	Delta signals per transcript chunk	Escalates while the conversation develops

OFFLINE GUARANTEE Intermediate chunks never require the LLM. The final path also falls back to rules-only analysis when the model is disabled, unavailable, or missing a key.

6. Privacy, Consent, and Safety

Privacy is an enforced data-flow rule rather than a claim. Text and supported speech modes are first analyzed locally. SAFE content creates no backend request. For SUSPICIOUS or CRITICAL results, the citizen reviews evidence and selects a scope before any network transmission.

Control	Implementation
Consent scopes	Selected evidence, redacted transcript, or full transcript
Local redaction	Masks Aadhaar-like, PAN-like, OTP/PIN, and account-like values
Retention	Citizen chooses 1, 7, or 30 days; expiry is stored and displayed
Withdrawal	Deletes the SQL incident, audit payload, and Neo4j report
Transparency	Privacy bar shows model, location, audio memory, and bytes shared
NCRP handoff	Opens the official search and copies an identifier locally, never in the URL

SAFETY BOUNDARY PRAHARI is decision support. Evidence packages require authorized human verification and are not legal determinations.

7. Graph Intelligence

Each consented risky report is modeled as a Report node. MERGE operations collapse repeated PhoneNumber, UPI_ID, BankAccount, and ClaimedAgency identifiers. Weakly Connected Components recover planted rings, while degree centrality ranks hub identifiers. New reports can join an existing ring immediately through a reused discriminative identifier.

Officer command workflow

1. Review incident KPIs and the risk-ranked queue.
2. Select a ring and inspect reports, cities, agencies, and its early-warning index.
3. Explore the graph to identify shared payment destinations and caller infrastructure.
4. Download the evidence package for human review and follow-up.

8. End-to-End Workflow

Step	Citizen/system action	Data movement
1	Capture live speech or paste suspicious text	Remains in browser memory
2	Run local phase and risk analysis	No evidence uploaded
3	Show critical intervention	No evidence uploaded
4	Citizen approves an evidence scope	Only the approved package is sent
5	Backend analyzes, audits, and persists	PostgreSQL incident and audit
6	Graph ingestion links shared identifiers	Neo4j report and identifiers
7	Officer dashboard highlights the ring	Ranked operational intelligence

9. Validation and Results

The prototype includes a reproducible 120-case synthetic English, Hindi, and Hinglish benchmark: 60 scams, 30 benign examples, and 30 hard negatives. These are prototype measurements, not field validation.

Metric	Result	Interpretation
Scam recall	90.0%	Detected nine in ten synthetic scams
Precision	100.0%	No false scam labels in the benchmark
F1 score	94.7	Strong precision/recall balance
Benign false-positive rate	0.0%	Benign examples stayed safe
Hard-negative critical rate	0.0%	RBI/UPI information did not become critical
Detected before payment language	90.0%	Most scams were recognized before the drain phrase

Integration evidence

- Live sessions verified from SAFE to SUSPICIOUS to CRITICAL.
- All five phases and highest phase 5 verified through the HTTP API.
- Consented final report verified with graph_synced=true and seeded RING-0 recovery.

- Caller number entered the graph while the citizen number remained excluded.
- Consent withdrawal removed the temporary incident and graph report.
- Frontend build, TypeScript, focused lint, and Python compilation passed.

10. Deployment and Demonstration

Docker Compose defines PostgreSQL, Neo4j with Graph Data Science, FastAPI, and Next.js. The seed script applies migrations, inserts synthetic incidents, initializes the graph, ingests reports, detects three planted rings, and prints the judge URLs.

Surface	URL	Purpose
Citizen check	localhost:3000/check	Live protection, privacy review, consent, and complaint draft
Officer command	localhost:3000/command	Queue, KPIs, graph, hubs, and rings
Backend API	localhost:8000/docs	Interactive API contract
Neo4j Browser	localhost:7474	Technical graph inspection

Three-minute judge narrative

1. Run the live call demo and show phase indicators update before the automatic stop screen.
2. Paste the seeded CBI transcript and emphasize that the verdict is private and 0 B has been shared.
3. Approve selected evidence and show the receipt, retention choice, and RING-0 join.
4. Open officer command, inspect the highlighted ring and hubs, then close on the citizen-to-intelligence loop.

11. Implemented, Synthetic, and Simulated Scope

Category	Included scope
Implemented	Web routes, local risk analysis, live transcript handling, consent, redaction, retention, deletion, persistence, audit, Neo4j ingestion, ring APIs, officer graph, evidence package
Synthetic	Seed incidents, planted rings, demo transcripts, and the 120-case benchmark
Simulated	Family Shield notification delivery
Not claimed complete	Image/PDF/audio-file extraction, carrier telephony integration, field accuracy, and multi-replica live sessions

12. Limitations and Next Steps

- Validate accuracy on independently collected, consented, multilingual field data.
- Move live-session state to an encrypted, expiring shared store for horizontal scaling.
- Add SIP/PSTN connectors only after explicit legal, consent, and telecom review.
- Expand private speech accuracy for low-resource languages and noisy calls.
- Integrate verified NCRP/1930 APIs if an authorized interface becomes available.
- Conduct accessibility, adversarial, privacy, and officer-workflow evaluations.

13. Team Contributions

Contributor	Primary ownership
Ivan	Platform, database, Docker, officer dashboard, integration, evidence package, load/offline checks, and release documentation
Mit	Deterministic rules, structured classifier, fusion, offline fallback, and evaluation harness
Adith	Citizen experience, timeline, intervention, Family Shield, language/explainability UI, and complaint export
John	Neo4j ingestion, ring detection, hub ranking, ring APIs, and planted-ring validation

Conclusion

PRAHARI demonstrates a credible intervention architecture for digital-arrest scams. Its strongest contribution is the privacy-preserving connection between live citizen protection and cross-report fraud intelligence. The prototype warns early, explains why, asks permission before sharing, and gives officers a connected view of repeated scam infrastructure.

FINAL POSITION Stop the scam before the money moves. Keep the citizen in control of their evidence. Turn approved signals into collective protection.

Appendix A. Key API Endpoints

Method	Endpoint	Purpose
POST	/api/analyze	Consent-gated analysis and graph sync
POST	/api/analyze/session/start	Begin an incremental live session
POST	/api/analyze/session/{id}/chunk	Analyze an intermediate or final chunk
GET	/api/rings	List ranked fraud rings
GET	/api/rings/{id}	Return reports, hubs, and graph JSON
GET	/api/rings/{id}/evidence-package	Return the human-review evidence package
DELETE	/api/incidents/{id}	Withdraw consent and delete linked records

Appendix B. Repository and Reporting Links

Repository: <https://github.com/ivanmathewsiby/PRAHARI>

National Cyber Crime Reporting Portal: <https://cybercrime.gov.in>

National Cyber Crime Helpline: 1930